

because of the amount of light it emitted, which was a first in itself, it came to be referred to as BOAT which stands for Brightest Of All Time. Not to be confused with GOAT or the Greatest Of All Time. The name will be even more justified after the presentation of a simple fact that is that this specific one was brighter by a factor of 10.

Officially nicknamed GRB 221009A, at initial checking it seems to be triggered by a supernova which, in turn, gave birth to a black hole. This event would have happened 2 billion light-years away from earth. And the fact that it was still so bright leads scientists to believe that the energy released equals roughly three suns converting all of the energy they



have in mass, to pure energy. Let that sink in!

It is not only the Burst that created a flurry among all astronomers. When space dust interacts with gamma-rays, there tends to be an afterglow that may appear longer than the blast itself. The Gemini South telescope, which is set up in Chile, is being used to explore this further. Considering that the light was so bright, telescopes will be able to see it for months to come furthering the research in this subject.

But is this a one-off event that occurred on its own or was it a series and there were others too? Let's delve into that next.

WHAT ELSE?

Tsvi Piran, an astrophysicist at the Hebrew University of Jerusalem,

While speaking to the Wired, Wen - fai Fong, an astrophysicist at Northwestern University said "I kept thinking, is this real? Because if it is, its an extremely rare, once in a lifetime type of event."

stated this about an incident that occurred recently, "This is the Rosetta Stone of Gamma-ray bursts".

He said this after a certain incident happened wherein there were two subsequent eruptions of gamma rays from stars in far-off galaxies that literally showered the earth with the highest ever energy photons.

The record for having the most number of electric photons in a Gamma-ray burst has been broken several times. The latest was in July 2018 wherein, in Namibia's capital,

Windhoek, photons from a GRB were recorded to have as many as 440 billion electron volts. The shocking part of this was that this was approximately 10 hours after the first burst.

Until this point, there was very little information about GRBs in general. Because there were two explosions, the shock waves that they produced accelerated them so much that they were able to travel at the speed of light. The photons gain energy from other photons in the galaxy. This process is referred to as inverse Compton scattering and is responsible for giving photons all of their energy.

This discovery was made only because of the proximity in which the blasts occurred and would have been unknown had that event not occurred the way it did.

While this may be a great event, and can this help us for our future, will this promote further researches in the subject?

WHAT DOES THE FUTURE HOLD FOR THEM?

A lot of researchers are excited by the amount of possibilities this once-in-a-century Gamma-ray Burst may have created for them. Firstly it was more recent and brighter. The brighter component helps a lot because it means that it will remain there for longer so it can be studied for a longer period, the work of which has already started. Such events are very helpful for scientists because they can be studied to know about origins of the universe and its heaviest elements. Scien-



tists can also decipher more about the secrets of the dark matter and maybe solve the mystery of how black holes form.

Other than that this specific event can be studied to know more about GRBs and maybe also find out why this one proved to be so intense that it was the brightest. Basically this can allow us to study the history of the universe and try to find out more secrets that the universe holds. **d**